The Key: Design Document

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Project Overview

The original shooting game has been completely redesigned and transformed into a rhythm-based music game. The project aims to combine entertainment with educational value.
The gameplay is presented using the traditional five-line musical staff, encouraging players to recognise pitch positions and understand basic music notation. The game supports two staves: treble clef staff and bass clef staff. Two players can perform simultaneously, each controlling one staff. Alternatively, a single player can control both registers at the same time, increasing the challenge and engagement. In Free Compose mode, the game records the sequence of notes played by the players. And in Song Performance mode, the system records the performance scores of players.

Genre Transformation

Instead of shooting enemies, the player now aligns with incoming musical marks and performs notes at the correct pitch and time. The transformation from shooter to rhythm game involved replacing core mechanics:
• The player character has been redesigned as a quarter note symbol.
• The original shooting key(space) has been transformed into a mute key for musical performance.
• Vertical movement now changes pitch position on the staff rather than aiming.
• Enemies have been replaced by scrolling musical marks that indicate required notes.

Game Modes

Mode 1: Free Compose
This mode allows for continuous performance without automatic termination. The player exits manually. Record the musical scores played by the players on each staff on the summary page.
Mode 2: Song Performance
This mode gives the prompt marks of twinkle twinkle little star, players should perform the correct notes as possible. This mode ends automatically when all musical marks leave the screen. The final scores for accuracy of performance are displayed for both players.

Key Achievements

Piano Simulation

The game simulates piano performance by requiring vertical pitch selection and timed triggering. Providing two player mode, players can coordinate both treble and bass parts similarly to two-hand piano playing.

Music Theory Details

Several music theory conventions were implemented.
• When the player mutes the music, displayed notes are converted into rests rather than simply removing sound.
• Stem direction follows engraving rules: Notes in the upper half of the staff have downward stems; Notes in the lower half have upward stems. This keeps stems visually centred and reinforces authentic notation principles.

Multiple Control Modes

The system supports:
• Two-player mode (each player controls one staff).
• Single-player two-staff mode.
Each register maintains independent scoring.

Game Event System

Mouse input is used for menu interaction.
During gameplay:
Treble Player
• W / S: Vertical pitch movement
• Space: Mute note
Bass Player
• Up / Down Arrow: Vertical pitch movement
• Right Arrow: Mute note
Global Controls
• R: Quit game
• D: Go to game summary

Implementation

The implementation is primarily based on Y-position mapping.
Two musical staffs are drawn first. The Y-coordinates of each staff line and the distance between lines determine the spacing between semitone steps.
• Each pitch corresponds to a fixed Y-position.
• Player movement is restricted to these discrete Y-values.
• An index value maps directly to both pitch and audio file.
Scrolling musical marks are generated as indexed sequences. In Song Performance mode, scoring is calculated when a mark reaches the fixed performance line at approximately $x = 700$. If the player's Y-position (pitch index) matches the incoming mark's index within the timing window, a score is awarded.
This replaces the original collision detection system with a pitch-time matching mechanism.

Materials and Asset Sources

All image assets were independently designed using ArtStudio.
All musical materials were independently recorded and generated using GarageBand.
The implementation is based on the original shooting game framework provided. The system was developed by modifying and extending the source code without using additional external libraries.